

RAGAS vs BLEU, ROUGE, and BERTScore

Detailed Study Notes

RAGAS, BLEU, ROUGE, and BERTScore are evaluation metrics commonly used in NLP and Generative AI. Traditional metrics (BLEU, ROUGE, BERTScore) compare generated text with a reference answer, while RAGAS is specifically designed for evaluating Retrieval-Augmented Generation (RAG) systems.

1. BLEU (Bilingual Evaluation Understudy)

Purpose: Measures similarity between generated text and reference text using n-gram overlap.

Used For: Machine Translation, Text Generation.

Formula Idea:

$BLEU = \text{Brevity Penalty} \times \text{Geometric Mean of n-gram Precisions}$

Example:

Reference: "The cat is sitting on the mat"

Prediction: "The cat sits on the mat"

BLEU checks matching word sequences.

2. ROUGE (Recall-Oriented Understudy for Gisting Evaluation)

Purpose: Measures overlap between generated text and reference text focusing on recall.

Common Types:

- ROUGE-1 (unigrams)
- ROUGE-2 (bigrams)
- ROUGE-L (longest common subsequence)

Example:

Reference: "AI is transforming healthcare."

Prediction: "AI transforms healthcare."

ROUGE evaluates how much important information is preserved.

3. BERTScore

Purpose: Uses contextual embeddings from BERT models.

Advantage: Captures semantic similarity rather than exact word matching.

Example:

Reference: "The student solved the problem."

Prediction: "The learner answered the question."

BLEU may score low, but BERTScore recognizes semantic similarity.

4. RAGAS (Retrieval-Augmented Generation Assessment)

Purpose: Evaluate Retrieval-Augmented Generation systems without requiring ground-truth references.

Main Metrics:

- Faithfulness
- Answer Relevancy
- Context Precision
- Context Recall
- Context Relevancy

Example:

Question: "Who invented Python?"

Retrieved Context: "Python was created by Guido van Rossum."

Generated Answer: "Python was created by Guido van Rossum in 1991."

RAGAS evaluates whether the answer is supported by retrieved context.

Comparison Table

Feature	BLEU	ROUGE	BERTScore	RAGAS
Needs Reference?	Yes	Yes	Yes	No (often)
Word Matching	High	High	Low	Low
Semantic Understanding	Low	Low	High	High
For Summarization	Limited	Excellent	Good	Not Primary
For Translation	Excellent	Moderate	Good	Not Primary
For RAG Systems	Weak	Weak	Moderate	Excellent
Context Evaluation	No	No	No	Yes

Simple Diagram

Traditional Evaluation

Reference Answer $\rightarrow$ Compare $\rightarrow$ BLEU / ROUGE / BERTScore

RAG Evaluation

Question → Retriever → Context → LLM → Answer

↓
RAGAS Metrics

Advantages and Disadvantages

Metric	Advantages	Limitations
BLEU	Simple and fast	Misses semantic meaning
ROUGE	Strong for summarization	Depends on reference overlap
BERTScore	Captures meaning	Computationally expensive
RAGAS	Designed for RAG pipelines	Requires LLM-based evaluation

Interview & Exam Points

1. BLEU focuses on precision of n-grams.
2. ROUGE focuses mainly on recall.
3. BERTScore uses contextual embeddings from transformer models.
4. RAGAS is specifically built for Retrieval-Augmented Generation evaluation.
5. RAGAS can evaluate faithfulness and retrieval quality.
6. Traditional metrics require reference answers; RAGAS often does not.
7. BERTScore handles paraphrasing better than BLEU and ROUGE.